

■ Team Playbook: Sepsis Digital Twin Capstone Project

1. ■ Data Handling Rules

- Source: Dataset.csv (hourly ICU data, ~546k rows, 44 cols).
- ID Structure: Use Patient_ID to group records. Always split data by patient, not by row, to avoid leakage.
- Splits: Train 70%, Validation 15%, Test 15% (by patient).
- Missing Data: LOCF for vitals, medians for labs, add mask vars.
- Normalization: Scale features (z-score/min-max), save scalers.

2. ■■ Model Development Standards

- Baselines: SIRS, qSOFA, NEWS, SOFA + Logistic Regression, RF, XGBoost.
- Advanced Models: LSTM, GRU-D, Transformer.
- Horizon: Predict sepsis 4–6h before onset.
- Evaluation: AUROC, AUPRC, Sensitivity @ 80% specificity, timeliness.

3. ■ Explainability & Trust

- Post-hoc Explainability: SHAP, LIME, Integrated Gradients.
- Top 5 contributing features per patient prediction.
- Report in clinical terms: Rising HR, Low MAP, High Lactate.

4. ■ Validation & Reporting

- Primary Dataset: Dataset.csv. External optional: PhysioNet 2019, eICU.
- Always compare baseline vs ML vs DL.
- Visualize ROC, PR, calibration, trajectories.
- Document preprocessing & hyperparams.

5. ■■■ Collaboration Workflow

- Version Control: GitHub branches per member.
- Code: Python (Pandas, Sklearn, PyTorch/TF, SHAP).
- Meet weekly, deliver notebooks + outputs.
- Shared doc for report, shared folder for figures.

6. ■ Milestone Expectations

- First Review (Aug): Dataset cleaned, baseline scoring, DL plan, dashboard mockup.
- Second Review (Sep/Oct): Baseline ready, DL results, SHAP prototypes.
- Final Review (Nov): Mature DL, explainability, dashboard, documentation.
- Final Report (Dec): IEEE paper, capstone report, demo video.

7. ■ Golden Rules

- Never mix patient records across train/test.
- Always compare against baselines.
- Check AUROC + PR + timeliness.
- Document all experiments.
- Accuracy + Explainability = Success.

■ Person-wise Roles and Responsibilities

■ Person A – Data Engineering & Management

Duties:

- Clean and preprocess dataset (handle NaNs, resample hourly).
- Perform feature engineering (rolling stats, SOFA components, shock index).
- Maintain consistent train/validation/test patient splits.
- Provide dataset documentation (stats, distributions, missingness analysis).

Deliverables:

- Final cleaned dataset with preprocessing scripts.
- Feature engineering notebook.
- Descriptive cohort statistics and visualizations.

■ Person B – Baselines & Clinical Scores

Duties:

- Implement clinical scoring systems (SIRS, qSOFA, NEWS, SOFA).
- Develop baseline ML models (Logistic Regression, Random Forest, XGBoost).
- Conduct ablation studies (vitals only, labs only).
- Benchmark models against clinical scores.

Deliverables:

- Baseline model performance tables (AUROC, AUPRC, Sensitivity, Specificity).
- Comparative analysis report.
- Figures: ROC/PR curves, calibration plots.

■ Person C – Deep Learning Models

Duties:

- Implement advanced models: GRU-D, LSTM, CNN-LSTM, Transformer.
- Train models for different prediction horizons (4–6h before onset).
- Calibrate predictions and measure timeliness.
- Document model architectures and hyperparameters.

Deliverables:

- Trained deep learning models and scripts.
- Risk trajectory plots and lead-time evaluation.
- Report section: Novelty and deep model performance.

■ Person D – Explainability & Dashboard

Duties:

- Apply explainability tools: SHAP, LIME, Integrated Gradients.
- Build Streamlit/FastAPI dashboard to simulate the digital twin.
- Integrate model predictions with interpretability outputs.
- Prepare presentation/demo materials.

Deliverables:

- Interactive digital twin dashboard.
- Explainability reports and feature attribution graphs.
- Demo video/screenshots for final presentation.